(28-08-26)

Friday, 28 August 2026 2:44 PM

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Folding → Shifting → Multi. → Add

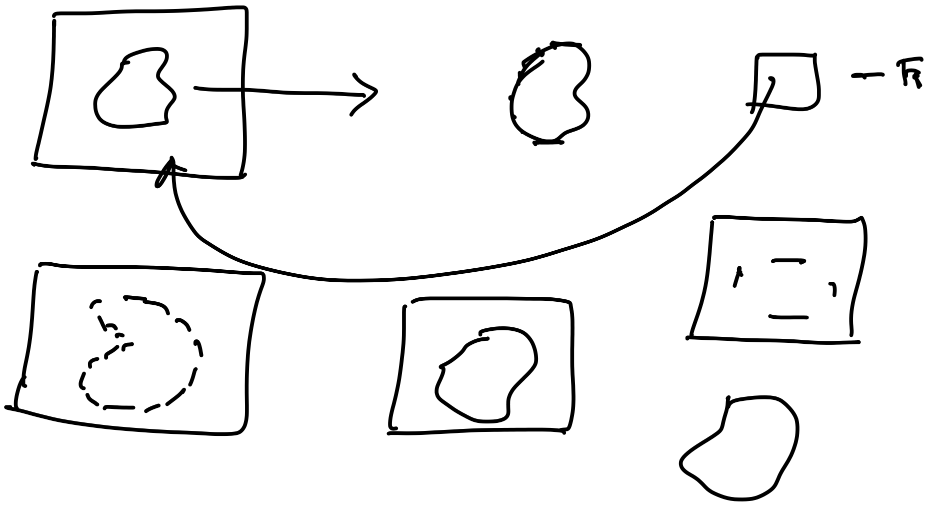
Convolution

Correlation

$$\boxed{y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(k-n)}$$

Auto

Cross



Some more properties

4) Identity ⇒ $x(n) * \delta(n) = x(n)$

5) Scalar Multiplication ⇒ $\boxed{a x(n)} * h(n) = a[x(n) * h(n)]$

6) Conv. with a α multiplier →

7) Time shifting ⇒ $x(n) * h(n) = y(n)$

⇒ $x(n-n_0) * h(n) = y(n-n_0)$

$x(n) * h(n-n_0) = y(n-n_0)$

8) Time reversal ⇒ $x(n) * h(n) = y(n)$

$x(-n) * h(-n) = y(-n)$

— x —

Properties of correlation

$$r_{x_1 x_2}(n) = \sum_{k=-\infty}^{\infty} x_1(k)\, x_2(k-n)$$

1) Even

$r_{x_1x_2}(l) = r_{x_1x_2}(-l)$ — Real signal

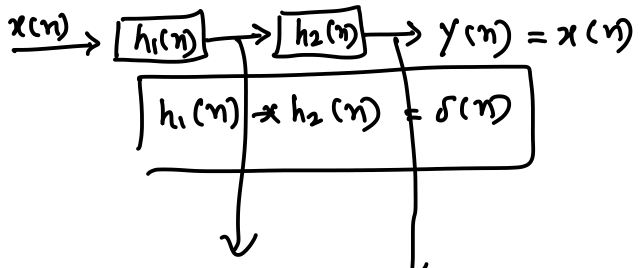
$r_{x_1x_2}(l) = r_{x_1x_2}^{*}(-l) \rightarrow$ Complex signal

2) Correlation with $\delta(n) \Rightarrow r_{x\delta}(l) = x(l)$

3) Relation with convolution $\Rightarrow r_{xy}(l) = x(l) * y(-l)$

— X —

Inverse System



$x(n) * h_1(n) * h_2(n) = x(n) * \delta(n) = x(n)$

— x —

Accumulator

$y(n) = \sum_{k=0}^{\infty} x(n-k)$ → Find inverse system ?

$h(n) = ?$

$h(n) = y(n) = \sum_{k=0}^{\infty} \delta(n-k) = u(n)$

$h'(n)$

$h'(n) * h(n) = \delta(n)$

$\underbrace{h'(n) * \sum_{k=0}^{\infty} \delta(n-k)} = \delta(n)$

$\Rightarrow \sum_{k=0}^{\infty} h'(n-k) = \delta(n)$

For $k = 0 \dots \infty$,

$\Rightarrow h'(n) + h'(n-1) + h'(n-2) \dots = \delta(n)$

$n = 0$

$\Rightarrow h'(0) + h'(-1) + h'(-2) \dots = \delta(0) = 1$

System is causal $\quad h'(n) = 0, \ n < 0$

$\Rightarrow h'(0) = 1$

$n=1$

$$\Rightarrow \quad h'(1) + h'(0) + h'(-1) + \cdots = 0$$

$$\Rightarrow \quad h'(1) + 1 + 0 + \cdots = 0$$

$$\Rightarrow \quad h'(1) = -1$$

$n=2$

$$\Rightarrow \quad h'(2) + h'(1) + h'(0) + \cdots = 0$$

$$\Rightarrow \quad h'(2) = 0$$

$$h'(3),\ h'(4) \cdots = 0$$

$$\boxed{h'(n) = [1, -1] = \delta(n) - \delta(n-1)}$$

Note Inverse system does not exist, the st is unstable

— x —

Upto this Midsem exam covers

— x —

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